Singidunum University

Soft Computing Optimization Techniques, 2025/2026

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Non-Aligned Movement
Imperialist Competitive Algorithm
(NAM-ICA)

Plain ICA with one rule changed: which empires are allowed to compete.

Early on, an empire competes only with its nearest rivals. The neighbourhood widens to global by the end.

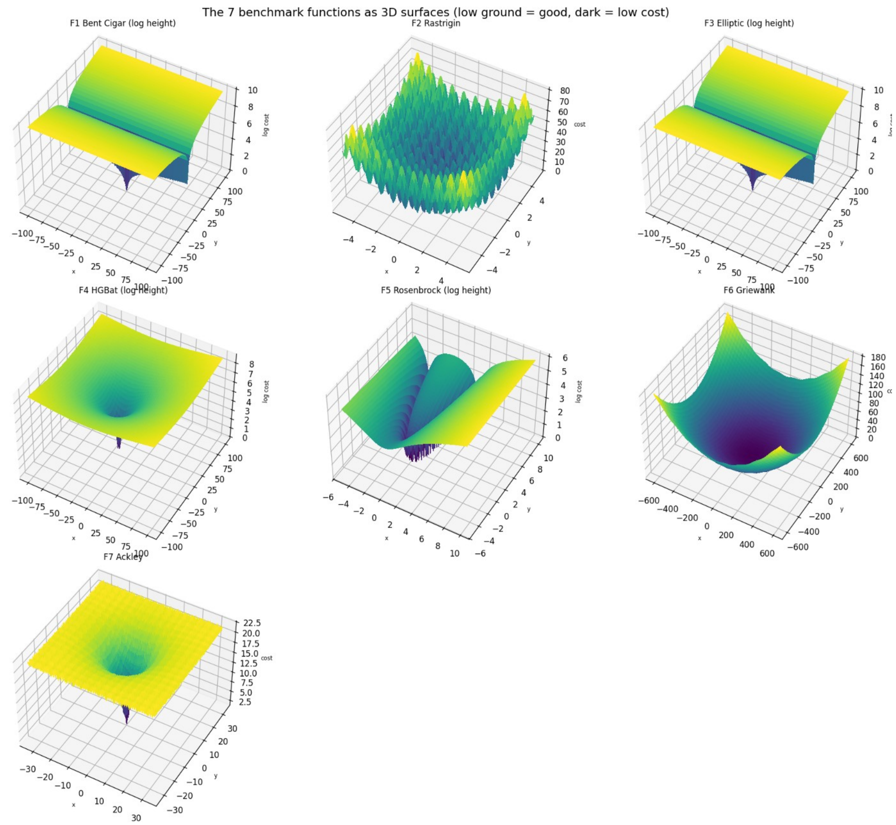
The problem: ICA gets stuck early

- ICA searches with competing empires. Strong empires absorb weak ones until one is left.
- Weakness: from the first round any empire can absorb any other, even a distant one.
- So the search commits to one region too early. This is called premature convergence.
- It hurts most on deceptive functions with many false low points, like Griewank and Ackley.

The idea: NAM-ICA in one sentence

- Change one rule: an empire may only take colonies from its k nearest rival empires.
- k starts at 2 (very local) and grows in a straight line to global by the final round.
- Local early keeps distant empires alive to explore. Global late still ends with one winner.
- No new tuned parameters: the neighbourhood grows automatically from local competition to global competition.

The seven benchmark functions



Low ground is good.

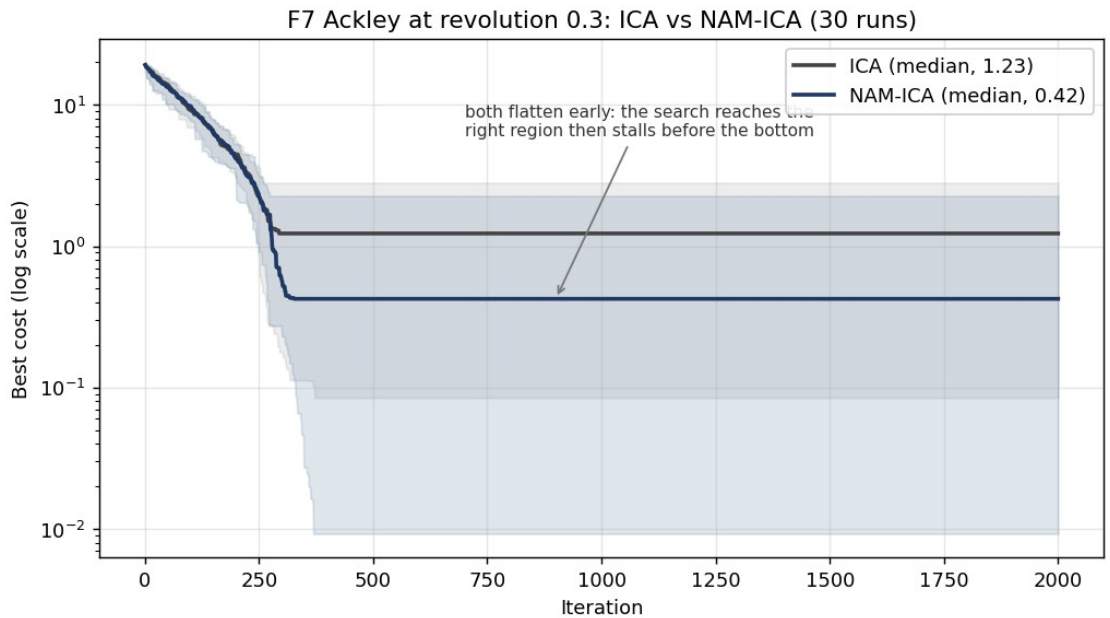
F1 and F3: thin skewed valleys. F2: bumpy bowl, still solved. F4: shifted bowl. F5: curved valley. F6: wide bowl with ripples. F7: flat plain with one hole.

Targets: F6 and F7

Setup and fairness

- 10 dimensions, 30 runs per function, 2000 iterations, 300 countries, 20 imperialists.
- Same fixed seeds 1 to 30 for every version, so the comparison is paired, run by run.
- Switch the rule off and the code reproduces plain ICA exactly. Same numbers, fair test.
- Four versions: ICA and NAM-ICA at revolution 0.3 (main) and 0.5 (higher randomness check).

Main result: Ackley (revolution 0.3)



Median cost dropped from 1.23 to 0.42.

This is a large median improvement, but p = 0.052, just above the usual 0.05 significance threshold

Both flatten near iteration 300: the search reaches the right region then stalls.

More iterations would not help.

The honest verdict

- Ackley: promising improvement, 1.23 to 0.42. The one real positive result.
- Griewank: no change. It was a target that did not work out, likely because its bowl already pulls every empire to the centre.
- HGBat at high randomness: NAM-ICA is much steadier ($p = 0.013$) than ICA. This is a secondary finding, not a planned aim.
- Everything else: ties.
- One simple rule, no new parameters. Modest, not a breakthrough.

Future work

- Add a local search step to push near-misses like Ackley closer to zero (costs extra evaluations).
- Test at higher dimensions, for example 30, to see if the benefit grows or shrinks.
- Run more seeds to confirm the results are stable.
- Start empires as real spatial clusters, so the regional separation exists from the start.
- Try other growth shapes for the competition neighbourhood.

Summary

- ICA can commit to one region too early.
- NAM-ICA changes who competes with whom: local first, global later.
- Ackley shows a promising median improvement.
- Griewank does not improve.
- HGBat gives a useful secondary result under high randomness.
- Overall: a small structural modification with modest but interesting results.

Thank you. Questions?